Indian Institute of Technology Kharagpur
Department of Mathematics
MA11003 - Advanced Calculus
Problem Sheet - 11

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1. Using Beta and Gamma functions prove the following:

   (a) $\displaystyle\int_0^\infty \sqrt{x}\; e^{-x^3} dx = \frac{\sqrt{\pi}}{3}$

   (b) $\displaystyle\int_0^\infty e^{-a^2x^2} dx = \frac{\sqrt{\pi}}{2a}$

   (c) $\displaystyle\int_0^1 x^3\; (1-x^2)^{\frac{5}{2}}\; dx = \frac{2}{63}$

   (d) $\displaystyle\int_0^{\frac{\pi}{2}} \sin^m x\; dx = \frac{\sqrt{\pi}}{2}\frac{\Gamma(\frac{m+1}{2})}{\Gamma(\frac{m+2}{2})}$

   (e) $\displaystyle\int_0^1 \sqrt{1-x^4}\; dx = \frac{\{\Gamma(\frac{1}{4})\}^2}{6\sqrt{2\pi}}$

   (f) $\displaystyle\int_0^{\frac{\pi}{2}} \sqrt{\tan x}\; dx = \frac{\pi}{\sqrt{2}}$

   (g) $\beta(m+1,n) = \frac{m}{m+n}\beta(m,n)$

   (h) $\displaystyle\int_a^b (x-a)^{m-1}(b-x)^{n-1} dx = (b-a)^{m+n-1}\beta(m,n)$

   (i) $\displaystyle\int_0^1 \frac{x^n}{\sqrt{1-x^2}}\; dx = \frac{\sqrt{\pi}}{2}\frac{\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n+2}{2})}$

   (j) $\displaystyle\int_0^1 x^{p-1}\; (1-x^r)^{q-1}\; dx = \frac{1}{r}\beta(\tfrac{p}{r},q)$

   (k) $\displaystyle\int_0^1 x^{p-1}(\ln \tfrac{1}{x})^{\alpha-1} dx = \frac{\Gamma(\alpha)}{p^\alpha}$

   (l) $\displaystyle\int_0^1 \frac{dx}{(1-x^n)^{\frac{1}{n}}}\; dx = \frac{1}{n}\Gamma(\tfrac{1}{n})\Gamma(1-\tfrac{1}{n})$

2. Given $\beta(x,y) = \displaystyle\int_0^1 t^{x-1}(1-t)^{y-1}dt,\; x>0,\; y>0$, show that

   (a) $\beta(x,y) = \displaystyle\int_0^{\frac{\pi}{2}} 2\sin^{2x-1}\theta\cos^{2y-1}\theta\; d\theta$

   (b) $\beta(x,y) = \displaystyle\int_0^\infty \frac{u^{x-1}}{(u+1)^{x+y}}du,\; x,\; y>0.$

   (c) $\beta(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$

   (d) $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

3. Show that

   (a) $\displaystyle\int_0^\infty x^m e^{-ax^n} dx = \frac{1}{n}a^{-\frac{m+1}{n}}\Gamma(\frac{m+1}{n})$, where m,n and a are positive integer .

   (b) $\displaystyle\int_0^1 x^m(\log\frac{1}{x})^n dx = \frac{\Gamma(n+1)}{(m+1)^{n+1}}$, where $m,n>-1$.

   (c) $\displaystyle\int_0^\infty x^m n^{-x} dx = \frac{m!}{(\log n)^{m+1}}$, where m is a non-negative integer and n is a positive constant.

4. Show that $\sqrt{\pi}\ \Gamma(2m+1) = 2^{2m}\Gamma(m+\frac{1}{2})\Gamma(m+1)$ for any positive integer m. Hence deduce that Legendre's duplication formula $\sqrt{\pi}\Gamma(2m) = 2^{2m-1}\Gamma(m)\Gamma(m+\frac{1}{2})$.

5. Given $\beta(n,\ 1-n) = \frac{\pi}{\sin n\pi}$ if $-1 < n < 1$, prove that $\displaystyle\int_0^1 \frac{x^n + x^{-n}}{1+x^2}dx = \frac{\pi}{2}\sec\frac{n\pi}{2},\ -1 < n < 1$.

6. Show that $\displaystyle\int_0^\infty \frac{x^m}{x^n+a}dx = \frac{1}{n}a^{\left(\frac{m+1}{n}-1\right)}\Gamma(\tfrac{m+1}{n})\Gamma(1-\tfrac{m+1}{n})$, where $a > 0$ and $0 < m+1 < n$.

7. Show that if m is a positive integer then

   (a) $2.\ 4.\ 6.\ 8.\ 10, ..., 2m = 2^{2m}\Gamma(m+1)$.

   (b) $1.\ 3.\ 5.\ 7.\ 9, ..., (2m-1) = \frac{2^{1-m}\Gamma(2m)}{\Gamma(m)}$

8. Evaluate the integral $\displaystyle\int_0^1 \frac{x^\alpha - 1}{\log x}\,dx$, $(\alpha > -1)$ by applying differentiating under the integral sign.

9. Using differentiation under integral sign prove the following:

   (i) $\displaystyle\int_{-\pi/2}^{\pi/2} \frac{log(1+b\sin x)}{\sin x}\,dx = \pi\sin^{-1}b$, where $|b| < 1$.

   (ii)Prove that $\displaystyle\int_0^\infty \frac{\tan^{-1}\alpha x\tan^{-1}\beta x}{x^2}\,dx = \frac{1}{2}\log\left[\frac{(\alpha+\beta)^{\alpha+\beta}}{\alpha^\alpha\beta^\beta}\right]$, $\alpha > 0,\ \beta > 0$.

   (iii) If $\alpha > 0,\ \beta > 0$, prove that $\displaystyle\int_0^{\pi/2} \log(\alpha\cos^2\theta + \beta\sin^2\theta)\,d\theta = \pi\log\frac{\sqrt{\alpha}+\sqrt{\beta}}{2}$.

10. Let $f(x,t) = (2x+t^3)^2$ then

    (i) find $\displaystyle\int_0^1 f(x,t)\,dx$.

    (ii) Prove that $\frac{d}{dt}\displaystyle\int_0^1 f(x,t)\,dx = \int_0^1 \frac{\partial}{\partial t}f(x,t)\,dx$.

11. (i) Define $f : \mathbb{R}^2 \to \mathbb{R}$ by

$$f(x,t) = \begin{cases} \frac{\sin xt}{t} & \text{if } t \neq 0 \\ x & \text{if } t = 0, \end{cases}$$

    find F', where $F(x) = \displaystyle\int_0^{\frac{\pi}{2}} f(x,t)\,dx$.

    (ii) Given $f : x \to \displaystyle\int_0^{x^2} \tan^{-1}\frac{t}{x^2}\,dt$, find f'.

12. For any real numbers x and t, let

$$f(x,t) = \begin{cases} \frac{xt^3}{(x^2+t^2)^2} & \text{if } x \neq 0, t \neq 0 \\ 0 & \text{if } x = 0, t = 0 \end{cases}$$

and $F(t) = \displaystyle\int_0^1 f(x,t)\,dx$. Is $\frac{d}{dt}\displaystyle\int_0^1 f(x,t)\,dx = \int_0^1 \frac{\partial}{\partial t} f(x,t)\,dx$? Give the justification.

13. Find the value of the integral $\displaystyle\int_0^\infty \frac{e^{-bx}\sin ax}{x}\,dx$, where $a > 0,\ b > 0$ are fixed, and hence deduce the value of the integral $\displaystyle\int_0^\infty \frac{\sin ax}{x}\,dx$.

14. Find the value of the following integrals

(i) $\displaystyle\int_0^{\frac{\pi}{2}} \log(1 - x^2\sin^2\theta)\,d\theta,\ |x| < 1$

(ii) $\displaystyle\int_0^\infty \frac{e^{-px}\cos qx - e^{-ax}\cos bx}{x}\,dx$

(iii) $\displaystyle\int_0^\infty e^{-x^2}\cos 2ax\,dx$